

Phase 3 Developer Guide

Files added

- ``src/pages/dashboard_page.py``
- ``src/pages/analytics_page.py``
- ``src/services/correlation_service.py``

Files replaced

- ``src/app.py``
- ``src/services/analytics_service.py``
- ``src/config.py``
- ``requirements.txt``

CorrelationService

``build_daily_dataset()`` converts four record sets into one date-indexed table.

``correlation_matrix()`` removes the date column, keeps metrics with at least two measurements, and calculates Pearson correlations.

``strongest_relationships()`` reads only the upper triangle of the matrix so each pair appears once. Pairs are ordered by absolute coefficient.

Adding a metric to correlations

1. Add the database and model field in Phase 2 code.
2. Add the field to the correct mapping in ``build_daily_dataset()``.
3. Choose ``_daily_sum()`` for additive daily measurements or ``_daily_mean()`` for observations.
4. Add a friendly label to ``METRIC_LABELS``.
5. Restart Streamlit and enter enough repeated data.

Percent changes

``percent_change_for_frame()`` sorts records chronologically before comparing the last two valid measurements. This avoids calculating changes from the descending

history order returned by repositories.